

Objective

This research aims to leverage machine learning interatomic potentials (MLIPs) to elucidate the multi-scale dynamics of battery electrolytes. By bridging the gap between microscopic molecular interactions and bulk macroscopic transport properties, this research seeks to establish a predictive framework for the rational design of high-performance electrolytes.^{1, 2}

Introduction

Sodium-ion batteries are a vital, low-cost alternative to lithium-ion systems, but performance hinges on electrolyte efficiency. Current research focuses on understanding single-solvent battery electrolyte mixtures, such as NaPF₆ in dimethoxyethane, to establish baseline solvation behaviors. The next frontier involves complex binary mixtures that can be tuned to balance high salt solubility with low viscosity. This research aims to bridge the gap between microscopic molecular interactions and macroscopic battery performance to design electrolytes that outperform traditional electrolyte mixtures

Methodology and Workflow

1. Molecular Optimization:

Individual solvent and salt geometries (NaPF₆, DME, PC) were pre-optimized using **Avogadro** to identify local energy minima.

2. System Assembly:

Optimized molecules were packed into 30 Å cubic cells at 1.0 M concentrations.

3. Simulation Engine:

Dynamics were driven by the UMA-OMol potential, which is specifically trained for battery electrolyte environments.

4. Equilibration:

- NVT Ensemble: Initial thermalization to 298.15 K.
- NPT Ensemble: Maintained at 1 bar to allow for volume fluctuations and equilibrium density reaching.

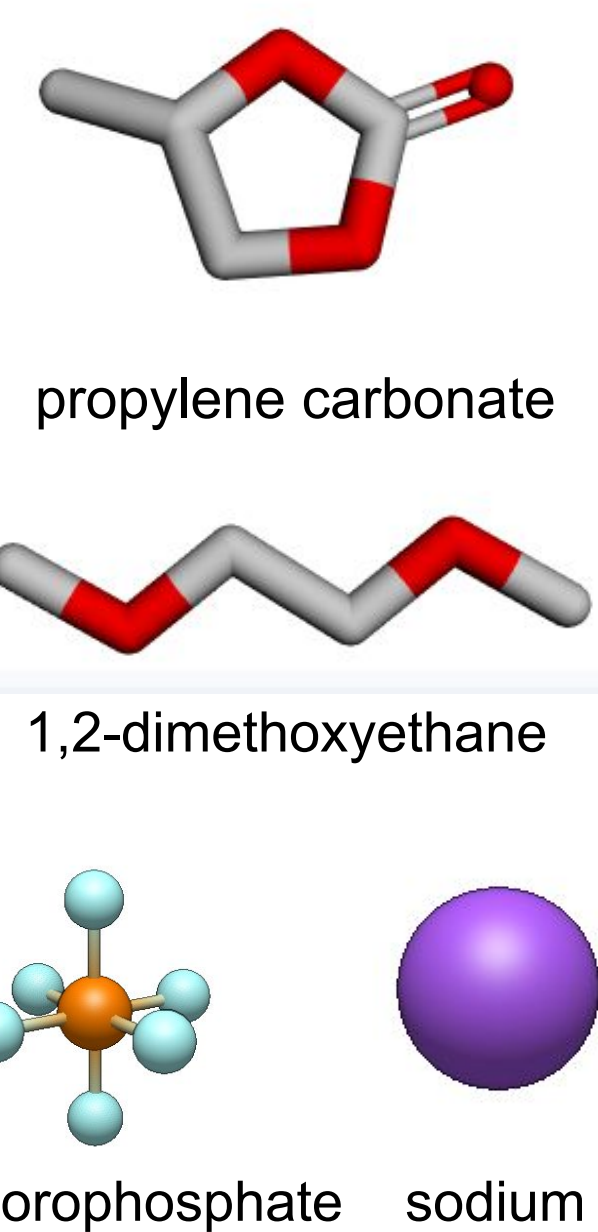
The Role of MLIPs

Traditional classical force fields often lack the chemical accuracy to describe complex ion pairing, while quantum mechanical methods are too slow for large-scale liquid systems.

We utilize Machine Learning Interatomic Potentials (MLIPs) to address this limitation.

What is MLIP?

MLIPs are neural networks trained on quantum mechanical data to predict atomic forces. They allow us to simulate thousands of atoms over long nanosecond timescales while maintaining ab initio accuracy. This enables the discovery of solvation structures that traditional models miss.



Mechanisms of Ion Solvation

Recent study by Kumar et al. showed that the solvation structure of Na ion battery electrolytes is highly sensitive to the topology of battery solvents, we are now studying how this solvation structure changes in the presence of mixtures. These are mixtures of 1,2-dimethoxyethane (DME) and propylene carbonate (PC). By systematically varying the composition of these binary systems, starting with a baseline of 1.0 M in pure DME, NaPF₆, we aim to quantify how the addition of a high dielectric carbonate (PC) alters the Na coordination environment. To establish a structural benchmark for this study, we constructed preliminary simulation cells (30 Angstrom cubic) representing the 1.0 M NaPF₆ in DME system, as visualized in the following simulation cells:

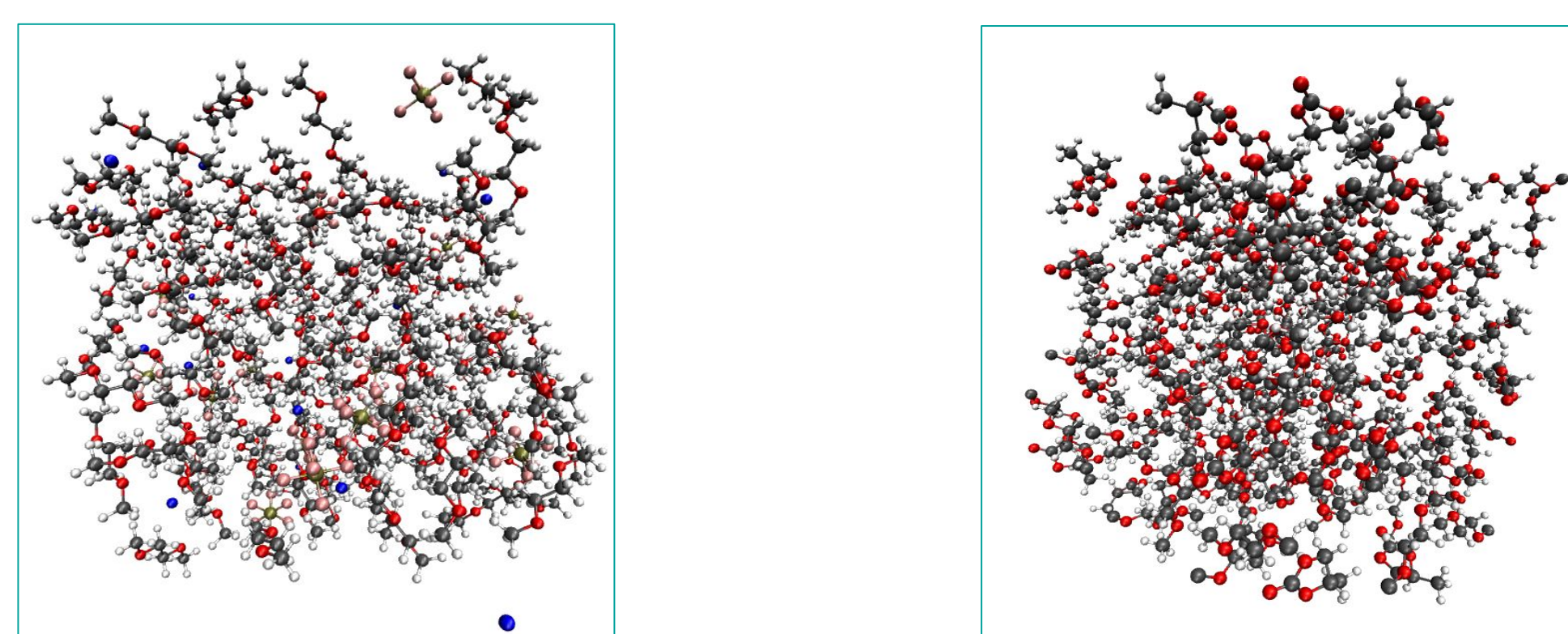


Fig. 1: Representative 1.0 M NaPF₆ simulation cells in pure DME (left) and binary DME/PC mixtures (right).

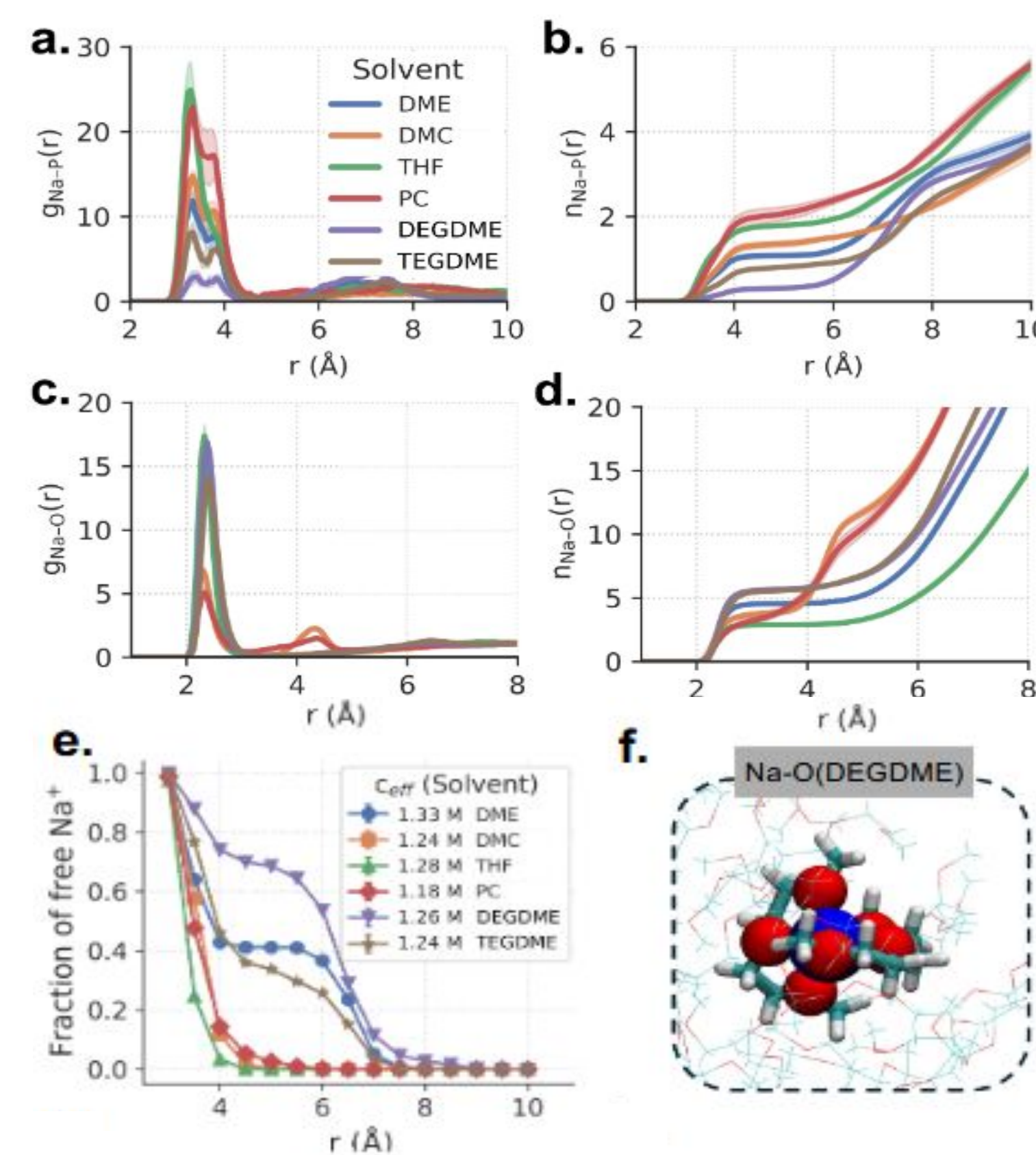
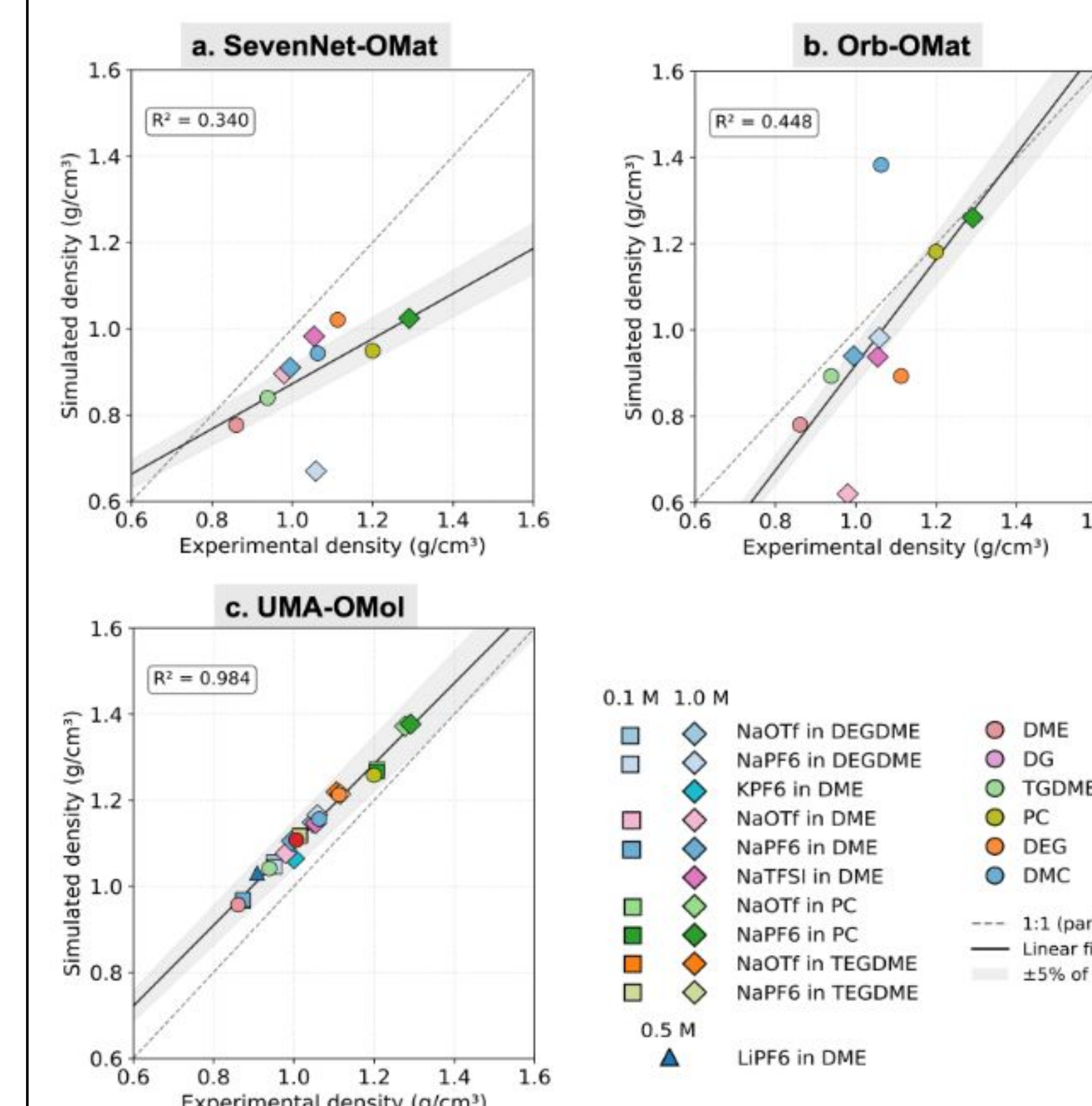


Fig. 2: Solvation Structure and Ion Association Metrics (Kumar et al.)

- (a–d): RDFs and coordination numbers establish the gold standard for Na⁺ interaction across different solvents.
- (e): Quantifies the transition from free ions to contact ion pairs (CIPs).
- (f): Close-up of the chelation mechanism in glyme-based systems

The above results show the sensitivity of sodium ion solvation structure on solvent topology. These benchmarks define the established structural and dynamic limits of sodium-ion solvation against which my DME and DME/PC simulation cells are directly compared. This comparison validates the UMA-OMol potential's performance in characterizing the specific ion-pairing and coordination shifts required to optimize electrolyte transport.

Model Validation & Benchmarking



While models trained on inorganic materials (SevenNet, Orb-OMat) show high scatter and low correlation (R^2), the UMA-OMol model achieves near-unity correlation ($R^2=0.984$), justifying its use for high-fidelity electrolyte simulations.¹

Future Direction

The next phase of this research involves molecular dynamics simulations-based screening of diverse solvent topologies and concentrations to optimize electrolyte performance. We will investigate:

- High-Throughput Screening: Testing varied solvent ratios (e.g., 80:20 vs 50:50) to optimize coordination shell stability.
- Computational Scaling: Leveraging NERSC (Perlmutter) supercomputing resources to accelerate large-scale discovery.
- Interface Dynamics: Extending simulations to the electrode interface to characterize the initial stages of SEI formation

Acknowledgements

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References

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2. Kumar, N., Rishko, W., Fiedler, K.R., Hollas, A., Chun, J. and Johnson, S.I., 2023. Correlations between Molecular Structure, Solvation Topology, and Transport Properties of Aqueous Organic Flow Battery Electrolyte Solutions. *ACS Materials Letters*, 5(11), pp.3050-3057.